UniVersHOI: Unifying Interaction Versatility and Scene Scalability for Cooperative Human-Object Interaction

Anonymous CVPR submission

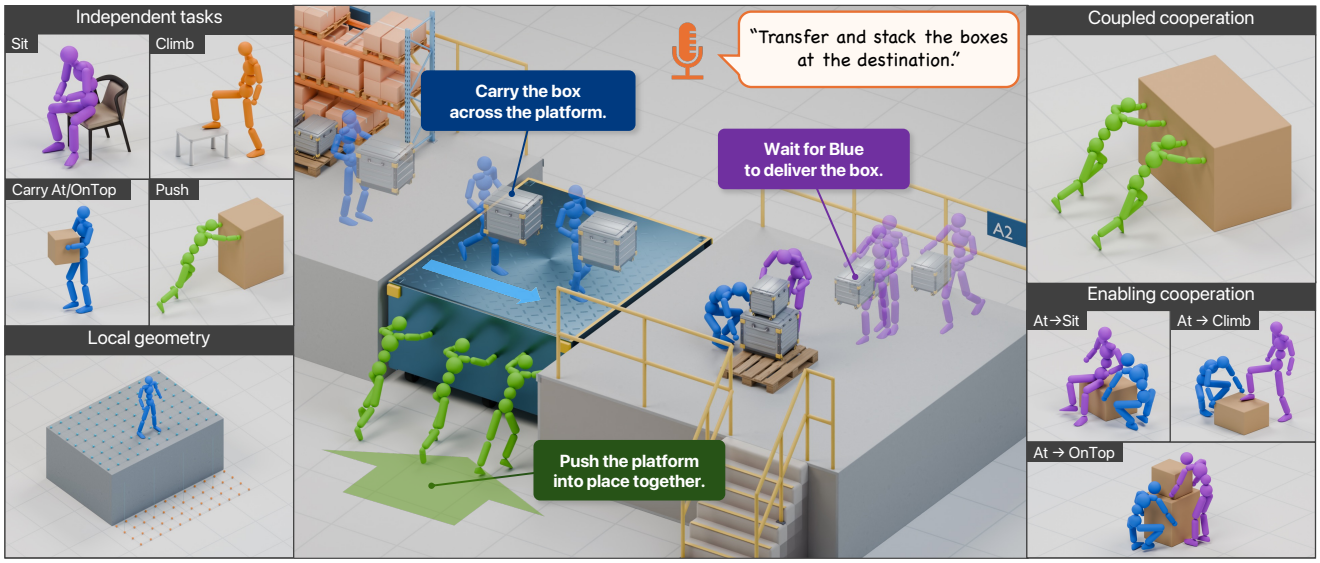


Figure 1. Overview of versatile cooperative HOI.

Abstract

Multi-agent physical interaction requires humanoid agents to perform diverse interaction skills and coordinate under different inter-agent dependencies across varying scene configurations. Existing approaches primarily focus on versatile individual interactions or specific cooperative behaviors, leaving their unified control across diverse scenes challenging. We introduce UniVersHOI, a unified physics-based policy that combines interaction versatility and scene scalability for cooperative human-object interaction. Our framework consists of two complementary components. First, a Scalable Scene-Centric Transformer (SCT) jointly encodes humans, objects, and goals using task-relation attention biases and task-action fusion, enabling task-specific control across varying scene configurations. Second, a Dependency-Aware Coordination Transformer (DCT) incorporates inter-agent dependencies into human self-attention and feature representations, enabling coordination of independent, enabling, and coupled interactions without manually specified low-level coordination phases. The resulting policy generates actions for all agents in a single forward pass. We evaluate interaction versatility and scene scalability across varying numbers of agents, including configurations beyond the training scale, and analyze coordination under different dependencies. We further integrate an LLM planner and local geometry observations to execute cooperative scenarios from high-level instructions.

1. Introduction

Physics-based Human-Object Interaction (HOI) is an important research area for modeling physical human behaviors in embodied AI and virtual character animation. Real-world environments involve varying numbers and configurations of people and objects. As these configurations change, interaction targets and cooperative partners also vary. Within these environments, people perform diverse physical interactions independently or in cooperation with others. Cooperation involves both jointly manipulating a shared object and coordinating different tasks through physical dependencies. As illustrated in Fig. 1, multiple agents jointly push a block to form a bridge, allowing another agent to cross and place an object for a follower to climb onto. Reproducing such environments with virtual humanoids requires coordinating diverse physical interactions and cooperative behaviors while accommodating varying agent and object configurations.

Recent advances in physics-based human interaction have enabled diverse individual HOI [2, 20, 36], physical Human-Human coordination [12, 23], and cooperative object manipulation [3, 16, 33]. However, existing approaches often focus on either diverse individual interactions or specific cooperative behaviors, leaving the unified control of diverse interaction skills under different inter-agent dependencies challenging.

Our goal is to develop a unified physics-based policy that executes diverse physical interactions across varying scene configurations while coordinating agents according to their inter-agent dependencies. Interaction versatility refers to the ability to perform diverse interaction skills under different dependency configurations. Scene scalability refers to the ability to apply the same policy across varying numbers and configurations of agents and objects. First, as scene compositions and task assignments vary, each agent needs to focus on information relevant to its assigned interaction. Second, different inter-agent dependencies require distinct forms of behavioral coordination, challenging a single policy to accommodate diverse cooperative interactions.

To address these challenges, we propose a unified physics-based policy comprising two complementary components: a Scalable Scene-Centric Transformer (SCT) and a Dependency-Aware Coordination Transformer (DCT). The SCT jointly encodes agents, objects, and goals using a variable number of entity tokens to accommodate different scene configurations. Task-relation attention biases guide information exchange among task-relevant entities, while task-action fusion directly conditions each agent's representation on its assigned skill. Building on these interaction-aware representations, the DCT incorporates dependency-relation attention biases to guide information exchange among agents according to their inter-agent dependencies. Dependency feature fusion directly incorporates dependency information into agent representations. The resulting policy simultaneously controls independent interactions, enabling cooperation between tasks, and coupled manipulation of shared objects within shared scenes, without manually specified low-level coordination phases.

We evaluate interaction versatility and scene scalability in scenes combining independent and cooperative interactions, using the same trained weights across scene sizes. Evaluations of enabling and coupled interactions assess cooperative behavior, and component ablations examine the contributions of the proposed mechanisms. For cooperative scenario execution, an LLM planner translates high-level instructions into task assignments and dependency relations. We adapt the policy with local geometry features to account for environmental constraints. The Bridge and Room scenarios combine joint manipulation, object delivery, and dependent interactions within a single execution sequence.

Our main contributions are summarized as follows:

- We propose a scalable scene-centric transformer that combines shared scene encoding, task-relation attention biases, and task-action fusion for multi-skill control across varying scene configurations.
- We introduce a dependency-aware coordination transformer that conditions inter-agent attention and agent representations on enabling and coupled dependencies for unified cooperative control.
- We demonstrate interaction versatility and scene scalability through unified control of diverse individual and cooperative interactions across varying scene sizes using the same policy.

2. Related Work

2.1. Versatile Human-Object Interaction

Physics-based human-object interaction has evolved from learning individual skills toward unified control of diverse behaviors [24, 25, 27]. PhysHOI [28] and InterMimic [35] learn contact-rich interactions through contact-aware imitation and teacher-student distillation, respectively. SkillMimic [29] learns reusable interaction skills, while SkillMimic-V2 [38] improves robustness and skill transitions through demonstration augmentation. UniHSI [32] and InterPhys [5] enable goal-directed scene interactions through language-guided contact planning and scene-conditioned adversarial imitation, respectively. TokenHSI [20] unifies diverse interaction skills through task tokenization, supporting flexible skill composition within a single policy. Recent works expand this versatility through VLM-guided interaction planning in VLM-RMD [2], generative control in InterPrior [36], and integrated scene interaction and object manipulation in Co²Skill [10] and CHIP [13]. However, these approaches primarily focus

on single-humanoid control without explicitly modeling inter-agent dependencies. In contrast, our method enables cooperative multi-humanoid interaction through explicit dependency-aware coordination.

2.2. Multi-Agent Interaction

Multi-agent interaction has been studied through both kinematic motion generation and physics-based character control. Kinematic approaches [4, 8, 14, 19] synthesize coordinated multi-human motions from language or observed partner motions. ReGenNet [34] models asymmetric action-reaction dynamics, while InterControl [30] uses pairwise joint-distance constraints to guide generation. Physics-based studies [9, 26, 31, 39] investigate multi-character interactions in simulation. InterAgent [12] employs an interaction-graph-based diffusion transformer for text-driven multi-agent control. AssistMimic [23] learns partner-aware tracking of contact-rich assistive motions through multi-agent reinforcement learning. While these studies advance motion coordination and physical interaction, coordinating diverse tasks under inter-agent dependencies remains challenging. Our method explicitly models these dependencies to enable versatile multi-agent physical cooperation within shared scenes.

2.3. Cooperative Human-Object Interaction

Cooperative motion modeling has advanced through interaction datasets and benchmarks [17], synchronized motion generation [6], stability-aware generation and contact refinement [11, 33], and interaction-preserving retargeting [15]. In physics-based control, Zhang et al. [39] reproduce cooperative carrying through interaction-graph-based imitation, while CooHOI [3] transfers individual interaction skills to cooperative control through shared object dynamics. TeamHOI [16] further supports varying team sizes through a decentralized policy with teammate tokens. Recent methods explore skill transfer and reuse in SynAgent [37] and From Solo to Ensemble [1], while Commander–Executor [7] combines high-level planning with physics-based control for cooperative object manipulation. These studies primarily focus on cooperation centered on shared-object manipulation. Our method explicitly models inter-agent dependencies across different tasks, where one agent's task enables another's, and coordinates these enabling interactions alongside joint object manipulation within a unified physics-based policy.

3. Method

3.1. Overview

Our goal is to develop a unified physics-based policy for multi-agent HOI that supports both interaction versatility and scene scalability. We denote the shared scene observation at time t by S_t, which contains information about N humanoid agents, objects, and goals. Task relations $\mathcal{R}_{\text{task}}$ specify each agent's interaction skill and its associated entities, while dependency relations $\mathcal{R}_{\text{dep}}$ describe enabling and coupled relationships among agents. Given these inputs, a single policy generates physical actions for all N agents:

$$(a_{1,t}, \ldots, a_{N,t}) = \pi_\theta(S_t, \mathcal{R}_{\text{task}}, \mathcal{R}_{\text{dep}}), \tag{1}$$

where each action $a_{i,t}$ specifies joint targets for the corresponding humanoid. As illustrated in Fig. 2, the Scalable Scene-Centric Transformer (SCT) jointly encodes scene entities with task relations, producing interaction-aware representations for each agent. The Dependency-Aware Coordination Transformer (DCT) then applies self-attention across these representations, conditioned on inter-agent dependency relations. A shared action head maps the resulting features to physical actions.

3.2. Scalable Scene-Centric Transformer

Shared Scene Representation. Given the shared scene observation S_t, we represent each humanoid agent, object, and goal entity u using a type-specific feature vector $\mathbf{x}_u$ and a pose T_u. The pose specifies the entity's position and orientation in a shared environment frame, mapping its local coordinate frame into that frame. Agent features encode articulated states, including body poses and velocities, while object features capture motion and bounding-box geometry. Goal entities use a constant feature, with target locations specified by their poses. Each feature vector $\mathbf{x}_u$ is encoded by a type-specific MLP and combined with a learnable type embedding to produce a 64-dimensional token. Both the encoders and type embeddings are shared across entities of the same type.

The resulting tokens are jointly processed by a shared transformer. To model geometric relationships across different token-local coordinate frames, we adopt Geometric Transform Attention (GTA) [18]. Using the entity poses T_u, GTA transforms queries, keys, values, and attention outputs to incorporate relative geometry between token-local frames, allowing each token to aggregate scene information in its own reference frame. Detailed transformations are provided in the supplementary material. The shared transformer produces representations for all agents in a single forward pass, without separately encoding the scene for each agent. Since the type-specific encoders and transformer parameters do not depend on entity counts, SCT supports scene scalability across varying numbers of agents, objects, and goals.

Task-Relation Attention Bias. We consider five interaction skills: SIT, CLIMB, AT (carrying and placing an object at a target location), ONTOP (carrying and placing an object onto another object), and PUSH (pushing an object

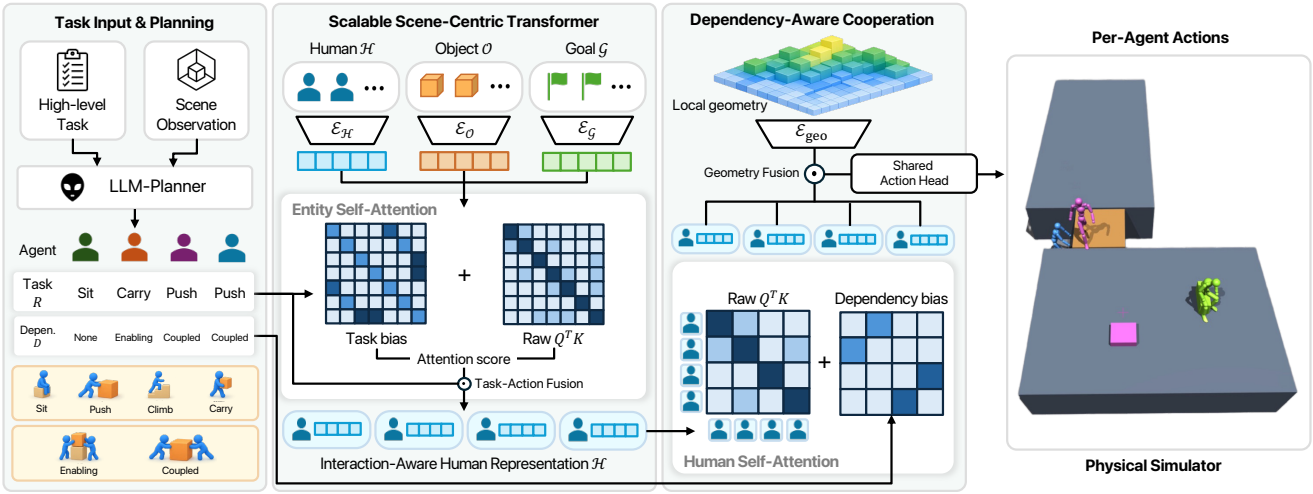


Figure 2. Overview of our proposed framework.

to a target location). Based on the task relations $\mathcal{R}_{\text{task}}$, we construct directed connections among the tokens associated with each agent's assigned interaction. For example, SIT connects an agent token to its supporting object, while AT establishes task connections from the agent to the manipulated object and goal, and from the object to the goal.

To condition attention on the assigned interaction skills, we compute a scalar bias for each task connection based on its corresponding skill s:

$$b_s = \mathbf{p}^\top \mathbf{e}_s. \tag{2}$$

Here, $\mathbf{e}_s$ is a learned skill embedding, and $\mathbf{p}$ is a learned projection vector specific to each attention head in each transformer layer. These biases form a matrix $\mathbf{B}_{\text{task}}$ according to the task connections, with zero bias for unconnected token pairs. The resulting matrix is added to the attention logits computed from GTA-transformed queries and keys:

$$\mathbf{A} = \text{softmax}\left(\frac{\widetilde{\mathbf{Q}}\widetilde{\mathbf{K}}^\top}{\sqrt{d}} + \mathbf{B}_{\text{task}}\right). \tag{3}$$

Here, $\widetilde{\mathbf{Q}}$ and $\widetilde{\mathbf{K}}$ denote GTA-transformed queries and keys, and d is the head dimension. All token pairs remain accessible through self-attention, allowing the policy to incorporate broader scene context.

Task-Action Fusion. While task-relation attention biases guide information exchange among scene entities, task-action fusion directly conditions each agent's representation on its assigned skill. At each transformer layer, the post-attention feature $\mathbf{z}_i$ of agent i is updated through a residual MLP:

$$\mathbf{z}_i \leftarrow \mathbf{z}_i + f_{\text{task}}\left([\mathbf{z}_i; \mathbf{e}_{s_i}]\right). \tag{4}$$

Here, $[\cdot;\cdot]$ denotes concatenation, $\mathbf{e}_{s_i}$ is the learned embedding of agent i's assigned skill, shared with the task-relation attention bias, and f_{task} is an MLP shared across agents and transformer layers.

We train the scene-centric policy on the five interaction skills in two-agent scenes using proximal policy optimization (PPO) [22] and an adversarial motion prior (AMP) [21]. Across all skills, task rewards follow a common structure comprising progress, state, and success terms, with skill-specific definitions. Detailed reward formulations and training settings are provided in the supplementary material. The resulting agent-token outputs form the interaction-aware representations $\mathbf{H} = [\mathbf{h}_1, \ldots, \mathbf{h}_N]^\top$, which serve as input to the dependency-aware coordination transformer.

3.3. Dependency-Aware Coordination Transformer

Dependency-Relation Attention Bias. Interaction versatility involves both diverse skills and coordination under different inter-agent dependencies. To model these dependencies, the DCT applies dependency-conditioned self-attention to the interaction-aware agent representations $\mathbf{H}$ produced by the SCT. We consider two cooperative dependency types, ENABLING and COUPLED. ENABLING connects a follower's query to its predecessor's key, while COUPLED establishes bidirectional connections among agents jointly manipulating an object. Agents performing independent interactions have no assigned inter-agent dependencies.

For each ENABLING or COUPLED connection, we compute a scalar attention bias from its corresponding relation embedding $\mathbf{e}_r$:

$$b_r = \mathbf{p}^\top \mathbf{e}_r. \tag{5}$$

Here, $\mathbf{e}_r$ is a learned relation embedding, and $\mathbf{p}$ is a DCT-specific projection vector learned separately for each layer

and attention head. These biases form a matrix $\mathbf{B}_{\text{dep}}$ according to the dependency relations, with zero bias for all other agent pairs. The resulting matrix is added to the agent self-attention logits:

$$\mathbf{A} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}} + \mathbf{B}_{\text{dep}}\right). \tag{6}$$

Here, $\mathbf{Q}$ and $\mathbf{K}$ are computed from the agent representations at the current layer, and d is the head dimension. All agent pairs remain accessible through self-attention, allowing information exchange beyond explicitly assigned dependencies.

Dependency Feature Fusion. In addition to dependency-relation attention biases, we directly condition each agent's attention context on its dependency type. At each transformer layer, the attention context of each ENABLING follower or COUPLED participant is updated through a residual MLP using the corresponding relation embedding:

$$\mathbf{m}_i \leftarrow \mathbf{m}_i + f_{\text{dep}}\left([\mathbf{m}_i; \mathbf{e}_r]\right). \tag{7}$$

Here, $\mathbf{m}_i$ denotes the attention-weighted value feature before the output projection, and $\mathbf{e}_r$ is the relation embedding shared with the dependency-relation attention bias. The MLP f_{dep} is shared across agents, relation types, and transformer layers. Other agents receive no dependency-specific fusion update. The resulting DCT outputs form coordinated agent representations $\mathbf{C} = [\mathbf{c}_1, \ldots, \mathbf{c}_N]^\top$, which are mapped to physical actions through a shared action head.

Enabling and Coupled Cooperation. For enabling cooperation, a predecessor places an object at a target location (AT), allowing a follower to perform SIT, CLIMB, or ON-TOP using the placed object. To guide the follower before the prerequisite is satisfied, we link the follower's dependent task to a virtual object token that retains the physical object's features and orientation but uses the predecessor's target position. This provides a spatial reference for preparation before the object is placed, while the DCT enables dependency-aware coordination.

For coupled cooperation, multiple agents jointly push a shared object. Each participant is assigned an object token corresponding to its pushing region, with all tokens reflecting the state of the same physical rigid body. Bidirectional COUPLED relations allow participants to coordinate their actions based on shared object dynamics. The variable-token design of SCT allows these cooperation-specific object representations to be incorporated into the shared scene without modifying the policy architecture.

Cooperative Policy Training. We train a unified cooperative policy across independent, enabling, and coupled interactions in four-agent scenes. The pretrained SCT actor encoder is frozen, while the DCT and shared action head are optimized using PPO [22] and AMP [21].

For enabling cooperation, the follower's progress reward remains active to encourage preparation, while state and success rewards for the dependent interaction require the predecessor's AT objective to be currently satisfied. After the follower enters a waiting region, a movement penalty encourages it to wait until the predecessor first satisfies its AT objective. For coupled PUSH, each agent receives rewards for approaching its assigned pushing region and moving the corresponding object region toward its goal. Each region is paired with a corresponding goal derived from the shared object's target position. The latter rewards are conditioned on all participants being positioned to push, encouraging them to act jointly on the shared object. Detailed reward formulations and training settings are provided in the supplementary material.

3.4. Cooperative Scenario Execution

To execute cooperative scenarios from high-level instructions, we employ an LLM planner that converts a task instruction and scene information into structured task assignments and inter-agent dependencies. The planner specifies each agent's interaction skill, associated object, and goal, forming the task relations and dependency relations used by our policy. While the planner determines the task structure, the learned policy generates physical actions without manually specified low-level coordination phases. Task and dependency assignments remain fixed throughout scenario execution.

To handle geometric constraints during execution, we incorporate local terrain observations into the cooperative controller. For each agent, we sample surface heights on an agent-centered, heading-aligned grid and encode the resulting height map using an MLP. The geometry feature is concatenated with the corresponding DCT representation and passed to a shared action head. We freeze the pretrained SCT and DCT and train the geometry encoder and geometry-conditioned action head for scenario execution. Detailed planner and geometry adaptation settings are provided in the supplementary material.

4. Experiments

We conduct experiments to evaluate whether a single physics-based policy can perform diverse interactions and coordinate agents under different inter-agent dependencies within shared scenes. We first evaluate interaction versatility across diverse skills and dependency configurations, together with scalability to varying scene sizes (Sec. 4.2), followed by a detailed analysis of enabling and coupled coordination (Sec. 4.3). We then examine the contributions of the proposed components (Sec. 4.4) and demonstrate integrated cooperative scenario execution (Sec. 4.5).

4.1. Experimental Setup

Implementation Details. We implement our framework in Isaac Gym, building upon the physics-based humanoid simulation environment of TokenHSI [20]. We adopt a goal-conditioned reinforcement learning framework, optimizing task-specific rewards using proximal policy optimization (PPO) [22] and adversarial motion priors (AMP) [21]. Reference motions are used for AMP and reference state initialization (RSI), without direct motion tracking or cooperative motion demonstrations. We first pretrain the Scalable Scene-Centric Transformer (SCT) on individual interaction skills in two-agent scenes. We then introduce the Dependency-Aware Coordination Transformer (DCT) and train a unified cooperative policy in four-agent scenes, sampling independent, enabling, and coupled interactions. During cooperative training, the pretrained SCT actor encoder is frozen, while the DCT and shared action head are optimized. Additional implementation details, motion data, and training configurations are provided in the supplementary material.

Baselines. We adapt TokenHSI [20] to our five interaction skills and shared multi-agent setting, denoted as TokenHSI*. For multi-agent execution, TokenHSI* applies its policy independently to each agent using agent-specific observations from the same shared scene. We also compare against the pretrained scene-centric policy (SCT), its cooperatively fine-tuned variant without the DCT (SCT + Coop. FT), and our full policy (Ours). Details of the baseline adaptations and training configurations are provided in the supplementary material.

Evaluation Metrics. We measure task completion using success rate (SR) with task-specific spatial criteria. For AT and PUSH, success requires the object center to be within 20 cm of its target in 3D space, consistent with the carrying-task evaluation in TokenHSI [20]. For SIT, CLIMB, and ONTOP, the humanoid pelvis or object center must lie within the central 80% of the supporting object's top surface along each horizontal dimension, with a vertical tolerance of 20 cm. We evaluate 512 scenes per setting across three seeds, with a maximum of 900 control steps (30 s) per episode, and report mean ± standard deviation under matched evaluation conditions.

We compute SR at the interaction-group level: independent interactions are evaluated per agent, enabling pairs succeed only when the predecessor's AT and follower's task conditions are simultaneously satisfied, and coupled groups succeed when their shared object reaches the goal. Each group contributes one outcome. For mixed scenes, we report the macro-average

$$\mathrm{SR}_{\mathrm{Mixed}} = \frac{\mathrm{SR}_I + \mathrm{SR}_E + \mathrm{SR}_C}{3}, \tag{8}$$

assigning equal weight to independent, enabling, and coupled interactions regardless of their sampling frequencies. Detailed evaluation settings are provided in the supplementary material.

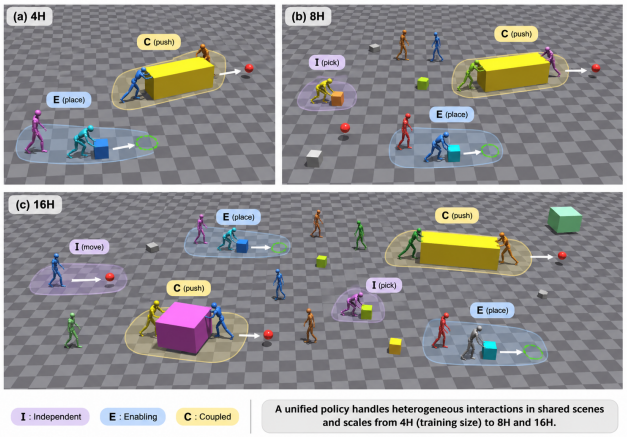


Figure 3. Illustrative mixed-scene execution with 4, 8, and 16 agents. Independent, enabling, and coupled interactions coexist within shared scenes of different sizes. **Temporary illustration; replace with actual simulation rollouts.**

4.2. Interaction Versatility and Scene Scalability

Tab. 1 evaluates interaction versatility and scene scalability across scenes with 2, 4, 8, and 16 agents. Independent scenes contain individually assigned interaction tasks, while mixed scenes comprise combinations of independent, enabling, and coupled interactions, as illustrated in Fig. 3. For each method, we use the same trained weights across all scene sizes without additional fine-tuning, including scenes with 8 and 16 agents beyond the cooperative training scale.

SCT achieves higher independent SR than TokenHSI*, while Ours further improves performance after cooperative training. Unlike TokenHSI*, which processes agents individually, SCT jointly encodes the shared scene and generates all agents' actions in a single forward pass. In mixed scenes, SCT + Coop. FT provides limited improvements over SCT, whereas Ours achieves substantially higher SR. These improvements extend to scenes with 8 and 16 agents.

4.3. Cooperative Interaction Performance

Tab. 2 evaluates enabling and coupled coordination under different inter-agent dependencies. For enabling interactions, we report Group SR and completion time for AT→SIT, AT→CLIMB, and AT→ONTOP, assigning the episode time limit to unsuccessful trials. For coupled interactions, we evaluate joint PUSH with 2, 3, and 4 participating agents using Group SR, final object-to-goal error, and a cooperative time ratio t_{coop}, adapted from TeamHOI [16]. The ratio measures the fraction of pushing time during which all participants simultaneously contact the shared ob-

Table 1. Interaction versatility and scene scalability. Independent (Ind.) and Mixed success rates (%) are reported as mean ± standard deviation across evaluation seeds.

	2H		4H		8H		16H	
Method	Ind.	Mixed	Ind.	Mixed	Ind.	Mixed	Ind.	Mixed
TokenHSI*	–	–	–	–	–	–	–	–
SCT	–	–	–	–	–	–	–	–
SCT + Coop. FT	–	–	–	–	–	–	–	–
Ours	–	–	–	–	–	–	–	–

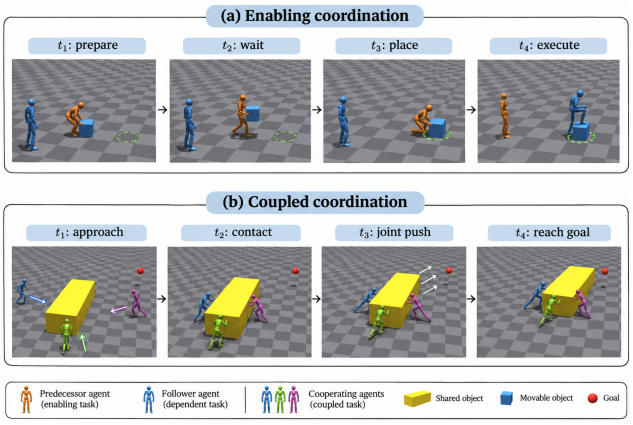


Figure 4. Illustrative sequences of dependency-aware physical coordination. (a) For enabling interaction, the follower prepares and adjusts its behavior while the predecessor performs AT, then executes the dependent task after valid placement. (b) For coupled interaction, agents approach assigned pushing regions, establish coordinated contact, and jointly manipulate the shared object. Frame labels describe behavior rather than externally prescribed control phases. **Temporary illustration; replace with actual simulation rollouts for the final paper.**

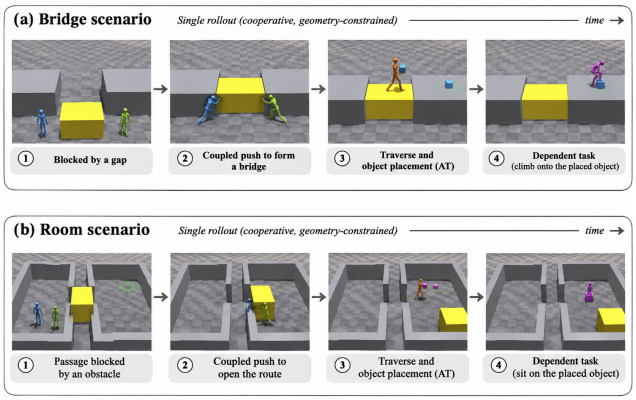


Figure 5. Cooperative scenario execution. (a) Bridge: joint PUSH establishes a traversable bridge, followed by object delivery and CLIMB. (b) Room: joint PUSH clears a passage, followed by object delivery and SIT.

ject. We additionally compare against SCT + FSM, which uses explicit waiting and release conditions with kinematic freezing, whereas our policy generates actions without manually specified low-level coordination phases.

Fig. 4 illustrates the resulting coordination behaviors. In enabling interactions, the follower prepares and waits while the predecessor places the required object, then executes its dependent task. In coupled interactions, multiple agents establish contact and jointly push a shared object toward its goal.

4.4. Ablation Study

Tab. 3 evaluates the contributions of the proposed SCT and DCT components. For SCT, we examine task-relation attention bias and task-action fusion using Independent SR across different scene sizes. For DCT, we evaluate dependency-relation attention bias and dependency feature fusion using Mixed SR across different scene sizes.

4.5. Cooperative Scenario Execution

Finally, we demonstrate integrated cooperative execution in two representative four-agent scenarios (Fig. 5). In both scenarios, two agents jointly push a shared object to establish a traversable path, while the other two agents perform an enabling interaction consisting of AT and a dependent task. In the Bridge scenario, joint PUSH forms a bridge across a gap, allowing an agent to deliver an object for another agent to climb onto. In the Room scenario, joint PUSH clears an obstructed passage, enabling object delivery followed by SIT. We incorporate local geometry features into the policy to support navigation through these environments. Both scenarios are executed using the same policy with fixed task and dependency assignments, without manually specified low-level coordination phases.

5. Conclusion

We presented UniVersHOI, a unified physics-based policy for cooperative human-object interaction that sup-

Table 2. Cooperative interaction performance. (a) Enabling Group SR (%) and completion time (s) for AT→SIT/CLIMB/ONTOP. (b) Joint-PUSH Group SR (%) for two, three, and four participants, final object-to-goal Error (cm), and cooperative time ratio t_{coop} (%). Higher is better except for Time and Error. Values are mean ± standard deviation across seeds.

(a) Enabling interaction

Method	AT→ SIT↑	AT→ CLIMB↑	AT→ ONTOP↑	Avg.↑	Time↓
TokenHSI*	–	–	–	–	–
SCT	–	–	–	–	–
SCT + FSM	–	–	–	–	–
Ours	–	–	–	–	–

(b) Coupled interaction

Method	2H SR↑	3H SR↑	4H SR↑	Error↓	t_{coop} ↑
TokenHSI*	–	–	–	–	–
SCT	–	–	–	–	–
SCT + FSM	–	–	–	–	–
Ours	–	–	–	–	–

Table 3. Component ablation. (a) SCT variants evaluated using Independent SR (%); (b) DCT variants evaluated using Mixed SR (%). † indicates inference-time removal; ‡ indicates retraining without the component. All values are mean ± standard deviation across evaluation seeds.

(a) Scene-centric interaction modeling

Method	1H↑	2H↑	4H↑	8H↑	16H↑
w/o Task-Relation Bias†	–	–	–	–	–
w/o Task-Action Fusion‡	–	–	–	–	–
SCT	–	–	–	–	–

(b) Dependency-aware coordination

Method	2H↑	4H↑	8H↑	16H↑
w/o Dependency-Relation Bias†	–	–	–	–
w/o Dependency Feature Fusion‡	–	–	–	–
Ours	–	–	–	–

ports diverse interaction skills and inter-agent dependencies across varying scene configurations. The Scalable Scene-Centric Transformer models task-relevant relationships among agents, objects, and goals, while the Dependency-Aware Coordination Transformer enables coordination according to inter-agent dependencies. Together, these components generate actions for all agents in a single forward pass, supporting independent, enabling, and coupled interactions without manually specified low-level coordination phases. We evaluate interaction versatility and scene scalability across varying scene sizes, including configurations beyond the cooperative training scale, and examine enabling and coupled coordination. Cooperative scenarios further demonstrate the integration of joint manipulation and dependent interactions with local geometry information. Extending the framework to broader forms of cooperative manipulation beyond joint pushing remains a direction for future work.

LLM Usage Statement

OpenAI

References

[1] Zekai Deng, Kangyi Chen, Ye Shi, and Jingya Wang. From solo to ensemble: A hierarchical framework for composable multi-agent human-object interaction. *arXiv preprint arXiv:2610.11722*, 2026. 3

[2] Zekai Deng, Ye Shi, Kaiyang Ji, Lan Xu, Shaoli Huang, and Jingya Wang. Human-object interaction via automatically designed VLM-guided motion policy. In *International Conference on Learning Representations (ICLR)*, 2026. 2

[3] Jiawei Gao, Ziqin Wang, Zeqi Xiao, Jingbo Wang, Tai Wang, Jinkun Cao, Xiaolin Hu, Si Liu, Jifeng Dai, and Jiangmiao Pang. CooHOI: Learning cooperative human-object interaction with manipulated object dynamics. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 79741–79763, 2024. 2, 3

[4] Anindita Ghosh, Rishabh Dabral, Vladislav Golyanik, Christian Theobalt, and Philipp Slusallek. ReMoS: 3D motion-conditioned reaction synthesis for two-person interactions. In *European Conference on Computer Vision (ECCV)*, 2024. 3

[5] Mohamed Hassan, Yunrong Guo, Tingwu Wang, Michael J. Black, Sanja Fidler, and Xue Bin Peng. Synthesizing physical character-scene interactions. In *ACM SIGGRAPH 2023 Conference Proceedings*, New York, NY, USA, 2023. Association for Computing Machinery. 2

[6] Wenkun He, Yun Liu, Ruitao Liu, and Li Yi. SyncDiff: Synchronized motion diffusion for multi-body human-object interaction synthesis. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 11731–11743, 2025. 3

[7] Buzhen Huang, Xingchen Wu, Yuanwang Yang, Chenyi Xie, and Kun Li. Scalable multi-human cooperative manipulation via commander-executor coordination. *arXiv preprint arXiv:2610.14911*, 2026. 3

[8] Muhammad Gohar Javed, Chuan Guo, Li Cheng, and Xingyu Li. InterMask: 3D human interaction generation via collaborative masked modeling. In *International Conference on Learning Representations (ICLR)*, pages 45683–45704, 2025. 3

[9] Kaiyang Ji, Ye Shi, Zichen Jin, Kangyi Chen, Lan Xu, Yuexin Ma, Jingyi Yu, and Jingya Wang. Towards immersive Human-X interaction: A real-time framework for physically plausible motion synthesis. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 10173–10183, 2025. 3

[10] Jeonghwan Kim, Hyeonwoo Kim, and Hanbyul Joo. Co^{2}Skill: Whole-body control via skill composition for long-horizon human-environment interaction. *arXiv preprint arXiv:2610.09291*, 2026. 2

[11] Jiye Lee, Yonghun Choi, and Jungdam Won. MOCHI: Motion enhancement of collaborative human-object interactions. *ACM Transactions on Graphics*, 45(4):1–18, 2026. 3

[12] Bin Li, Ruichi Zhang, Han Liang, Jingyan Zhang, Juze Zhang, Xin Chen, Lan Xu, Jingyi Yu, and Jingya Wang. InterAgent: Physics-based multi-agent command execution via diffusion on interaction graphs. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15253–15265, 2026. 2, 3

[13] Jianan Li, Xiao Chen, and Tien-Tsin Wong. Unifying physics-based humanoid interaction with a context-conditioned interaction prior. In *SIGGRAPH Asia 2026 Conference Papers*, 2026. 2

[14] Han Liang, Wenqian Zhang, Wenxuan Li, Jingyi Yu, and Lan Xu. InterGen: Diffusion-based multi-human motion generation under complex interactions. *International Journal of Computer Vision*, 132(9):3463–3483, 2024. 3

[15] Ziyu Lin, Shihang Du, Tianhang Wang, Xiaodong Liu, Runze Zhang, Zongyou Yu, Haodong Zhang, Alois Knoll, and Guang Chen. CoCarry: Contact-support-structured retargeting for dual-humanoid cooperative carrying, 2026. arXiv:2610.32460. 3

[16] Stefan Lionar and Gim Hee Lee. TeamHOI: Learning a unified policy for cooperative human-object interactions with any team size. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 37121–37132, 2026. 2, 3, 6

[17] Yun Liu, Chengwen Zhang, Ruofan Xing, Bingda Tang, Bowen Yang, and Li Yi. CORE4D: A 4D human-object-human interaction dataset for collaborative object REarrangement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1769–1782, 2025. 3

[18] Takeru Miyato, Bernhard Jaeger, Max Welling, and Andreas Geiger. GTA: A geometry-aware attention mechanism for multi-view transformers. In *International Conference on Learning Representations*, 2024. 3

[19] Sakuya Ota, Qing Yu, Kent Fujiwara, Satoshi Ikehata, and Ikuro Sato. PINO: Person-interaction noise optimization for long-duration and customizable motion generation of arbitrary-sized groups. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 10676–10685, 2025. 3

[20] Liang Pan, Zeshi Yang, Zhiyang Dou, Wenjia Wang, Buzhen Huang, Bo Dai, Taku Komura, and Jingbo Wang. TokenHSI: Unified synthesis of physical human-scene interactions through task tokenization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5379–5391, 2025. 2, 6

[21] Xue Bin Peng, Ze Ma, Pieter Abbeel, Sergey Levine, and Angjoo Kanazawa. AMP: Adversarial motion priors for stylized physics-based character control. *ACM Transactions on Graphics*, 40(4):144:1–144:20, 2021. 4, 5, 6

[22] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017. 4, 5, 6

[23] Yuto Shibata, Kashu Yamazaki, Lalit Jayanti, Yoshimitsu Aoki, Mariko Isogawa, and Katerina Fragkiadaki. Learning to assist: Physics-grounded human-human control via multi-agent reinforcement learning. In *Proceedings of the*

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. 2, 3

[24] Chen Tessler, Yunrong Guo, Ofir Nabati, Gal Chechik, and Xue Bin Peng. MaskedMimic: Unified physics-based character control through masked motion inpainting. *ACM Transactions on Graphics*, 43(6):1–21, 2024. 2

[25] Guy Tevet, Sigal Raab, Setareh Cohan, Daniele Reda, Zhengyi Luo, Xue Bin Peng, Amit Haim Bermano, and Michiel van de Panne. CLoSD: Closing the loop between simulation and diffusion for multi-task character control. In *The Thirteenth International Conference on Learning Representations*, 2025. 2

[26] Jiashun Wang, Jessica Hodgins, and Jungdam Won. Strategy and skill learning for physics-based table tennis animation. In *ACM SIGGRAPH 2024 Conference Papers*, pages 1–11, 2024. 3

[27] Wenjia Wang, Liang Pan, Zhiyang Dou, Jidong Mei, Zhouyingcheng Liao, Yuke Lou, Yifan Wu, Lei Yang, Jingbo Wang, and Taku Komura. SIMS: Simulating stylized human-scene interactions with retrieval-augmented script generation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 14117–14127, 2025. 2

[28] Yinhuai Wang, Jing Lin, Ailing Zeng, Zhengyi Luo, Jian Zhang, and Lei Zhang. Physhoi: Physics-based imitation of dynamic human-object interaction. *arXiv preprint arXiv:2312.04393*, 2023. 2

[29] Yinhuai Wang, Qihan Zhao, Runyi Yu, Hok Wai Tsui, Ailing Zeng, Jing Lin, Zhengyi Luo, Jiwen Yu, Xiu Li, Qifeng Chen, Jian Zhang, Lei Zhang, and Ping Tan. Skillmimic: Learning basketball interaction skills from demonstrations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17540–17549, 2025. 2

[30] Zhenzhi Wang, Jingbo Wang, Yixuan Li, Dahua Lin, and Bo Dai. InterControl: Zero-shot human interaction generation by controlling every joint. In *Advances in Neural Information Processing Systems*, pages 105397–105424, 2024. 3

[31] Jungdam Won, Deepak Gopinath, and Jessica Hodgins. Control strategies for physically simulated characters performing two-player competitive sports. *ACM Transactions on Graphics*, 40(4):1–11, 2021. 3

[32] Zeqi Xiao, Tai Wang, Jingbo Wang, Jinkun Cao, Wenwei Zhang, Bo Dai, Dahua Lin, and Jiangmiao Pang. Unified human-scene interaction via prompted chain-of-contacts. In *International Conference on Learning Representations*, 2024. 2

[33] Jiahao Xu, Xiaohan Yuan, Xingchen Wu, Chongyang Xu, Kun Li, and Buzhen Huang. Stability-driven motion generation for object-guided human-human co-manipulation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 38355–38365, 2026. 2, 3

[34] Liang Xu, Yizhou Zhou, Yichao Yan, Xin Jin, Wenhan Zhu, Fengyun Rao, Xiaokang Yang, and Wenjun Zeng. ReGenNet: Towards human action-reaction synthesis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1759–1769, 2024. 3

[35] Sirui Xu, Hung Yu Ling, Yu-Xiong Wang, and Liang-Yan Gui. Intermimic: Towards universal whole-body control for physics-based human-object interactions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12266–12277, 2025. 2

[36] Sirui Xu, Samuel Schulter, Morteza Ziyadi, Xialin He, Xiaohan Fei, Yu-Xiong Wang, and Liang-Yan Gui. InterPrior: Scaling generative control for physics-based human-object interactions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 23516–23527, 2026. 2

[37] Wei Yao, Haohan Ma, Hongwen Zhang, Yunlian Sun, Liangjun Xing, Zhile Yang, Yuanjun Guo, Yebin Liu, and Jinhui Tang. Synagent: Generalizable cooperative humanoid manipulation via solo-to-cooperative agent synergy. *arXiv preprint arXiv:2604.18557*, 2026. 3

[38] Runyi Yu, Yinhuai Wang, Qihan Zhao, Hok Wai Tsui, Jingbo Wang, Ping Tan, and Qifeng Chen. SkillMimic-V2: Learning robust and generalizable interaction skills from sparse and noisy demonstrations. In *ACM SIGGRAPH 2025 Conference Papers*, pages 1–11, 2025. 2

[39] Yunbo Zhang, Deepak Gopinath, Yuting Ye, Jessica Hodgins, Greg Turk, and Jungdam Won. Simulation and retargeting of complex multi-character interactions. In *ACM SIGGRAPH 2023 Conference Proceedings*, pages 1–11, 2023. 3